

# Applications and Implementation

## A LINEAR PROGRAMMING APPROACH TO THE DISCRIMINANT PROBLEM

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### ABSTRACT

We propose an alternative solution to the discriminant problem, one that requires little more than a minimum familiarity with linear programming. The approach shows promise for eliminating the complexities of conventional statistical approaches without sacrificing the essential power of existing methods.

*Subject Areas: Statistical Techniques, Linear Programming, and Linear Statistical Models.*

### INTRODUCTION

A variety of statistical problems require the capacity to assemble data into meaningful groups. Finding key characteristics necessary to differentiate like-valued data groupings is thus a common task of traditional statistical analysis. Cluster, classification, and discriminant analysis represent alternative approaches to conveniently identifying significant subsets of individual observations (entities) by exploring common and contrasting features. The methodologies developed have proved useful in a wide range of applications including medical research, archeological discovery, employee evaluation, and credit screening. (By establishing meaningful criteria for group differentiation, the decision maker might identify a likely cause for a set of symptoms, assess the origin of an ancient artifact, accurately predict the performance of a job applicant, or profile unacceptable credit risks.)

Unfortunately, the theoretical underpinnings of conventional statistical methods have often proved cumbersome, confusing, and, in some cases, inappropriate to the solution of such problems. We propose an alternative approach, one based on simple logic and elementary linear programming technique, that can be effectively used to reduce the complexities of current approaches without sacrificing the essential power of existing methods.

### BACKGROUND

Previous efforts to apply linear programming techniques to typically statistical problems have for the most part been restricted to regression-type procedures which have sought to produce minimum absolute deviation (M.A.D. or L1-Norm) solutions to compete with traditional least squares analysis [5] [9] [11] [14]. With origins in the early work of Charnes, Cooper, and Ferguson [4] and that of Charnes and Cooper [2], substantial research [1] [7] [8] [13] [15] has

generally validated the approach but to date has failed to induce widespread acceptance.

Of more relevance to our own efforts is Kendall's [10] discussion of an LP-related convex-hull procedure for establishing group membership. Although the method is well conceived, application appears somewhat limited. Rao [12] outlines an extensive set of linear and nonlinear integer programming formulations for the clustering problem, but admits to serious computational difficulties.

While these antecedent efforts have not always been entirely successful, they do suggest the appeal and potential of alternative LP-based approaches to problem types generally conceded to conventional statistics. Our goal is to clarify this potential with a simple formulation for the discriminant problem that offers useful advantages over existing alternatives.

### AN EXAMPLE

The following example, which typifies the problem of group discrimination, will be used to illustrate our method. Suppose we want to predict the performance of job applicants by finding some set of explanatory factors, revealed at the time of application, that when properly used, effectively distinguish successful from unsuccessful workers. Although we could conceivably use a broad range of factors to project potential performance, we will, for simplicity, use only two. We assume that these two characteristics can be measured adequately at the time of application and that such measurements can be taken for workers already hired.

Typically, there are four stages in the analysis of such a problem: (1) evaluate current and past employees as either successful or unsuccessful, (2) measure each employee on the selected characteristics (here, two), (3) compute a characteristic score for each employee which somehow combines these measurements—a score that one hopes can be used as a criterion to separate the successful from the unsuccessful groups, and (4) apply the scoring procedure to new job applicants in order to assign each applicant to the appropriate group. The object is to hire those whom we could confidently assign to the successful group and reject those who would project as unsuccessful.

While the preceding four-step procedure might be usefully probed on a number of counts, we will focus principally on the third step: determining an effective way to combine the set of individual measurements to produce an aggregate score that can be used in differentiating group membership. More specifically, we should like to identify a weighting scheme that, when applied to the measured values for an employee (current, past, or prospective), will produce a single value indicating the appropriate group assignment—either successful or unsuccessful. Implicit in such a strategy is the need to establish a critical value, or cutoff point, that will serve as the breakpoint between successful and unsuccessful scores. Thus we would like to be in a position to assert that a weighted score above (below) the critical value will assign an individual to the successful (unsuccessful) group.

Since there are a number of possible weighting schemes, the problem becomes one of establishing an appropriate criterion for judging what constitutes a

“good” (or even optimal) scheme. Certainly one would like to identify a scheme that would produce scores for employees already identified as successful which are far different from the scores that would be produced for unsuccessful employees. Thus, if we plan to use high scores to indicate successful and low scores to indicate unsuccessful employees, we would like the scheme to assign scores *far above* the critical value to successful employees and scores *far below* the critical value to unsuccessful employees. Such a scheme would ensure a sharp distinction between groups and promote confidence in the assignment of applicants to one of the two categories.

The procedure we propose directly incorporates just such a criterion. We will weight each of the measurements,  $x_{ij}$ , by some  $w_i$  so that, insofar as possible, scores produced for successful employees will be well above the critical value,  $c$ . Scores for unsuccessful employees will be well below  $c$ . In fact we seek to identify the set of weights  $w_i$  that will *maximize* the distance of any employee score from the critical  $c$ . Symbolically, we want to find  $w_i$ s such that

$$w_1x_{1j} + w_2x_{2j} \geq c \quad \text{for each successful employee } j, \quad (1)$$

$$w_1x_{1j} + w_2x_{2j} \leq c \quad \text{for each unsuccessful employee } j. \quad (2)$$

By introducing  $d$  into the expressions above to denote distance from the break-point  $c$ , we produce

$$w_1x_{1j} + w_2x_{2j} - d \geq c \quad \text{for each successful employee } j, \quad (3)$$

$$w_1x_{1j} + w_2x_{2j} + d \leq c \quad \text{for each unsuccessful employee } j. \quad (4)$$

Setting as our objective the maximization of this distance measure  $d$ , we produce a completed general model in the linear programming format

$$\text{Maximize } d \quad (5)$$

Subject to

$$w_1x_{1j} + w_2x_{2j} - d \geq c \quad \text{for each successful employee } j$$

$$w_1x_{1j} + w_2x_{2j} + d \leq c \quad \text{for each unsuccessful employee } j$$

$$(w_1, w_2, d \text{ unrestricted in sign}).$$

Suppose data for five successful and five unsuccessful employees are available. That is, suppose measurements have been taken on two relevant characteristics (years of prior experience and years of post-high school education, for example) for each of ten employees. Such data are displayed below.

| Successful Group |                   |                   | Unsuccessful Group |                   |                   |
|------------------|-------------------|-------------------|--------------------|-------------------|-------------------|
| Employee #       | Characteristic #1 | Characteristic #2 | Employee #         | Characteristic #1 | Characteristic #2 |
| 1                | 4                 | 1                 | 6                  | 1                 | 1                 |
| 2                | 5                 | 2                 | 7                  | 3                 | 1                 |
| 3                | 7                 | 2                 | 8                  | 2                 | 2                 |
| 4                | 9                 | 1                 | 9                  | 6                 | 3                 |
| 5                | 8                 | 4                 | 10                 | 3                 | 3                 |

We seek a weighting scheme to differentiate, insofar as possible, the two groups. To accomplish this the general LP procedure developed above can be adapted as follows:

Maximize  $d$

(6)

Subject to

$4w_1 + 1w_2 - d \geq 10$

$5w_1 + 2w_2 - d \geq 10$

$7w_1 + 2w_2 - d \geq 10$

$9w_1 + 1w_2 - d \geq 10$

$8w_1 + 4w_2 - d \geq 10$

Successful Group Constraints

$1w_1 + 1w_2 + d \leq 10$

$3w_1 + 1w_2 + d \leq 10$

$2w_1 + 2w_2 + d \leq 10$

$6w_1 + 3w_2 + d \leq 10$

$3w_1 + 3w_2 + d \leq 10$

Unsuccessful Group Constraints

$w_1, w_2, d$  unrestricted in sign.

The cutoff value of  $c$  has been arbitrarily set to 10, specifying that we seek a weighting scheme that will assign values greater than 10 to successful employees and values less than 10 to unsuccessful employees. (Actually, any positive value for  $c$  would be acceptable. Selection of a particular  $c$  value, while arbitrary, affects the scale of the result. Once  $c$  is chosen, its value should remain constant to allow comparison among alternative solution possibilities.)

The solution to the above LP problem yields weights  $w_1 = 5.71$  and  $w_2 = 8.57$ , which in turn would compute employee scores as follows:

| <u>Employee #</u> | <u>Score</u> |
|-------------------|--------------|
| 1                 | 14.3         |
| 2                 | 11.4         |
| 3                 | 22.8         |
| 4                 | 42.8         |
| 5                 | 11.4         |
| 6                 | -2.9         |
| 7                 | 8.6          |
| 8                 | -5.7         |
| 9                 | 8.6          |
| 10                | 8.6          |

The closest any employee score comes to the breakpoint 10 is 1.4 units. That is,  $d$  is here assigned a value of 1.4.

Since  $d$  is positive, the method has produced a weighting scheme which completely separates the employee groups. While the absolute magnitude of  $d$  is tied to our selection of the breakpoint value (here 10), the value of  $d$  relative to the range of employee scores would suggest a substantial gap between the two groups and thus effective discrimination. Had the value of  $d$  been much smaller with respect to the range of employee scores, a less powerful discriminant effect would be evident. Some additional observations are necessary. It seems clear that there will be situations in which it will simply not be possible to find a weighting scheme (at least not a linear weighting scheme) that will ensure perfect group separation. That is, we may well find that no matter how individual measurements are weighted to score group members, one or more individuals is misclassified: some successful group score(s) will fall below  $c$ , while some unsuccessful group score(s) will exceed the breakpoint. By allowing the value of  $d$  to be unrestricted in sign, we ensure that if some overlap of scores is in fact inescapable—forcing a negative  $d$ —the overlap will be *minimized* by the method. In the event that a large number of scores result in misclassification, additional explanatory characteristics may be added in an effort to produce better group discrimination.

Variations and fine tuning of the basic model are certainly possible. As proposed, the model uses a single distance variable,  $d$ . In maximizing  $d$ , we have in fact *maximized* the *minimum* distance of any group member's score from the breakpoint. It would be possible, and perhaps desirable, to assign a distance variable  $d_j$  ( $j = 1, \dots, m$ ) to each group member— $d_j$  measuring the distance of group member  $j$ 's score from breakpoint  $c$ . The objective might then be restated as:

$$\text{Maximize } \sum_{j=1}^m d_j \quad (7)$$

Subject to

$$w_1x_{1j} + w_2x_{2j} - d_j \geq c \quad \text{for each successful employee } j$$

$$w_1x_{1j} + w_2x_{2j} + d_j \leq c \quad \text{for each unsuccessful employee } j$$

( $w_1, w_2, d_j$  unrestricted in sign).

The solution to this LP problem would be a weighting scheme that would be less influenced by the minimum distance of any group score from the cutoff than by the aggregate distance of all the scores, which would tend to push the mass of the group back from the critical point. More complex extensions are considered in [6].

### SUMMARY

We have proposed an alternative solution to the discriminant problem, one that requires little more than a minimum familiarity with linear programming. Discussion has been restricted to the model in its simplest form, although more elaborate variations are certainly possible. It is our hope that more attention will be given to unconventional approaches to multivariate statistical problems as viable and attractive alternatives to the sometimes unwieldy statistical procedures currently available. [Received: January 16, 1978. Accepted: August 5, 1980.]

### REFERENCES

- [1] Ashar, V. G., & Wallace, T. D. A sampling study of minimum absolute deviation estimators. *Operations Research*, 1963, 11, 747-758.
- [2] Charnes, A., & Cooper, W. W. *Management models and industrial application of linear programming* (Vol. 1). New York: Wiley, 1961.
- [3] Charnes, A., & Cooper, W. W. Goal programming and multiple objective optimizations. *European Journal of Operational Research*, 1977, 1, 39-54.
- [4] Charnes, A., Cooper, W. W., & Ferguson, R. O. Optimal estimation of executive compensation by linear programming. *Management Science*, 1955, 1, 138-151.
- [5] Eisenbeis, F. A., & Avery, R. B. Two aspects of investigating group differences in linear discriminant analysis. *Decision Sciences*, 1978, 9, 487-493.
- [6] Freed, N., & Glover, N. *Discriminant analysis by the separating/goal hyperplane method*. Unpublished manuscript, University of Colorado, Management and Information Science Report Series, in preparation.
- [7] Glahe, F. R. The small sample properties of simultaneous equation least absolute estimators vis-a-vis least squares estimators. *Econometrics*, 1970, 38, 742-753.
- [8] Hunt, J. G., Dowling, J. M., & Glahe, F. R.  $L_1$  estimation in small samples with Laplace error distributions. *Decision Sciences*, 1974, 5, 22-29.
- [9] Karst, O. J. Linear curve fitting using least deviations. *Journal of the American Statistical Association*, 1958, 53, 118-132.
- [10] Kendall, M. G. Discrimination and classification. In P. R. Krishnaiah (Ed.), *Multivariate analysis*. New York: Academic Press, 1966.
- [11] Kiountouzis, E. A. Linear programming techniques in regression analysis. *Applied Statistics*, 1973, 1, 799-821.

- [12] Rao, M. R. Cluster analysis and mathematical programming. *Journal of the American Statistical Association*, 1971, 66, 622-626.
- [13] Usow, K. H. On  $L_1$  approximation: Computation for discrete functions and discretization effect. *SIAM Journal of Numerical Analysis*, 1967, 4, 233-244.
- [14] Wagner, H. M. Linear programming techniques for regression analysis. *Journal of the American Statistical Association*, 1959, 54, 206-212.
- [15] Wilson, H. G. Least squares versus minimum absolute deviations estimation in linear models. *Decision Sciences*, 1978, 9, 322-335.